Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Black, Franklin Owen M.D.		
Project Title:	Tactile Sensory Supplementation of Gravitoinertial References to Optimize Sensorimotor Recovery		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	fob@neurotology.org	**Fax:**	FY 503-413-5348
PI Organization Type:	NON-PROFIT	**Phone:**	503-413-5353
Organization Name:	Legacy Health System		
PI Address 1:	1225 NE 2nd Ave		
PI Address 2:	Suite 380		
PI Web Page:			
City:	Portland	**State:**	OR
Zip Code:	97232-2003	**Congressional District:**	3
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):	Wood, Scott (NASA JSC)		
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Task Description:	During the first year we addressed our first aim: postural control performance in standing subjects while providing orientation cues via tongue electrotactile display units (TDU) using sensors attached to different body segments (modified from original NRA proposal Specific Aim 1). Computerized dynamic posturography was used to measure postural sway stability with and without tactile displays with eyes closed during unexpected movements of the support surface. Twelve healthy subjects participated. Key findings: As hypothesized, improvements in postural stability were observed, as evidenced by reductions in peak-to-peak AP sway, when using the TDU. The largest improvements in postural stability occurred during the translated sway-referenced mode with or without support surface rotation. In this case, the peak-to-peak sway was significantly reduced (improved postural sway stability) with the accelerometers mounted on the torso compared to sensors mounted on subjects' head. Impact of findings: Locating the sensors on the torso instead of the head may be advantageous for controlling or improving postural equilibrium under conditions that		

	require head-on-trunk motion. Results of this experiment suggest that the effectiveness of tongue tactile sensory supplementation for countermeasure applications may be optimized by accounting for the body segment orientation needed for specific postural task performance, including g-transitions and changing support surface conditions. Proposed plan for coming year: Following additional data analysis, the above results will be submitted for publication. During the second year we will address the second aim: evaluate the efficacy of gravitoinertial reference displays to improve gaze stabilization during ambulation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The BrainPort tactile displays is currently being used in clinical trials for balance rehabilitation in patients with bilateral vestibular loss. This research is primarily targeted for NASA medical research and operational community to evaluate the feasibility of sensory tactile aid to improve postural and locomotion control. The results are also of use to clinical and military researchers interested in the use of tactile prostheses as balance, orientation or navigation aids. Results of this study will be of interest to vestibular rehabilitation professionals.
Task Progress:	The purpose of the first specifc aim was to compare postural control performance when providing orientation cues via tongue tactile displays using sensors attached either to the head or torso. We hypothesized that tactile sensory supplementation derived from sensors attached to the head would optimally improve standing postural performance by stabilizing the head in space. Postural equilibrium was measured with a computerized hydraulic platform in 12 healthy adults (7M, 5F, 25-63 y). Trials (100 s duration with eyes closed) were conducted with the support surface fixed, sway-referenced, during sum-of-sines perturbations (0.01, 0.145, 0.295 & 0.565 Hz, peak 2°/s) or during a combination of sway-referencing and sum-of-sines perturbations. Sway-referencing was accomplished by rotating the platform equal to AP sway angle, by translating the support surface proportional to AP sway, or by rotating and translating the support surface simultaneously. The linear accelerometers were either mounted on a head band or on a torso belt. Subjects were required to keep the intraoral display in their mouths on all trials, including trials that did not provide tactile feedback. As expected, there were improvements in postural stability, based on reduced peak-to-peak AP sway amplitude, when using the TDU. The largest improvements in postural control occurred with the platform in the translate sway-referenced mode with or without support surface rotation. The peak-to-peak sway was significantly reduced with the accelerometers (sensors) mounted on the torso compared to the head. We conclude that locating sensors on the torso may be advantageous for control postural equilibrium under conditions that naturally involve head-on-trunk motion, such as occurs with counter-rotation of the head during fore-aft support surface translational movements. These results suggest that the effectiveness of tactile sensory supplementation for balance prosthesis and vestibular rehabilitation applications may be optimized by accounting for the body segment orientation needed for specific task performance. An abstract summarizing these findings was accepted for presentation at the Seventh Symposium on the Role of the Vestibular Organs in Space Exploration in June 7-9, 2006. We have expanded these measurements using pseudorandom galvanic stimulation (GVS) to transiently destabilze postural control in normals. In collaboration with MacDougall and Moore (EBR, 2006) we demonstrated that pseudorandom GVS elicited similar postural decrements that occur in short-duration, orbital space flight returning crewmembers. Thus, pseudorandom GVS provides a ground-based model for the development of the TDU as a sensorimotor countermeasure for changes in g-transition and support surface conditions. Looking to the future, we have developed treadmill, dynamic visual acuity and OptoTrak motion analysis techniques to support our second year specific aim studies.
Bibliography Type:	Description: (Last Updated: 07/26/2016)
Articles in Peer-reviewed Journals	MacDougall HG, Moore ST, Curthoys IS, Black FO. "Modeling postural instability with Galvanic vestibular stimulation." Exp Brain Res. 2006 Jun;172(2):208-20. Epub 2006 Jan 24. PMID: 16432695 , Jun-2006